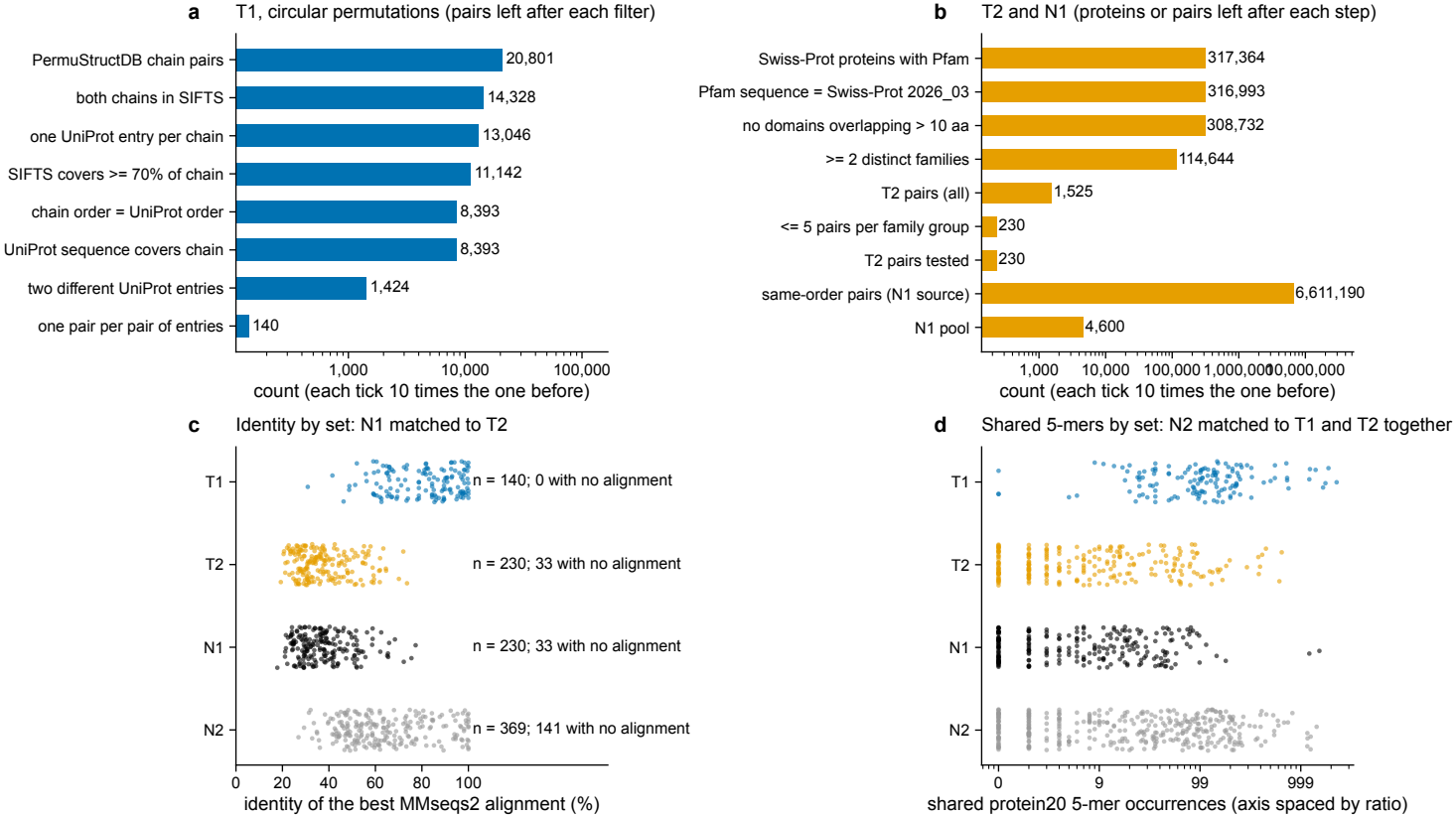

a
T1, circular permutations (pairs left after each filter)
PermuStructDB chain pairs
20,801
both chains in SIFTS
14,328
one UniProt entry per chain
13,046
SIFTS covers >= 70% of chain
11,142
chain order = UniProt order
8,393
UniProt sequence covers chain
8,393
two different UniProt entries
1,424
one pair per pair of entries
140
1,000
10,000
100,000
count (each tick 10 times the one before)
b
T2 and N1 (proteins or pairs left after each step)
Swiss-Prot proteins with Pfam
317,364
Pfam sequence = Swiss-Prot 2026_03
316,993
no domains overlapping > 10 aa
308,732
>= 2 distinct families
114,644
T2 pairs (all)
1,525
<= 5 pairs per family group
230
T2 pairs tested
230
same-order pairs (N1 source)
6,611,190
N1 pool
4,600
1,000
10,000
100,000
1,000,000
10,000,000
count (each tick 10 times the one before)
c
Identity by set: N1 matched to T2
T1
T2
N1
N2
n = 140; 0 with no alignment
n = 230; 33 with no alignment
n = 230; 33 with no alignment
n = 369; 141 with no alignment
0
20
40
60
80
100
identity of the best MMseqs2 alignment (%)
d
Shared 5-mers by set: N2 matched to T1 and T2 together
T1
T2
N1
N2
0
9
99
999
shared protein20 5-mer occurrences (axis spaced by ratio)